Controlled Class-Imbalance Construction for TCGA-UT

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Abstract

Class imbalance in computational pathology is not only a question of how many examples a class contributes. A class may perform poorly because it is rare, because its frozen foundation-model features overlap with other classes, or because both effects coincide. The companion TCGA-UT imbalance benchmark established the dataset, the mitigation taxonomy, and patch-level and WSI-bag evaluations under the native TCGA-UT long tail. This report defines a controlled counterpart: artificial full-scale TCGA-UT training distributions that keep all 32 cancer types, preserve fixed validation and test splits, and vary only the training support assigned to each class. Power-law training distributions are generated under native-prevalence class ordering at imbalance parameters $\lambda \in \{0.8, 1.1, 1.3\}$ with a fixed pool of 1,528 slides, yielding realized head-to-tail ratios of approximately 15:1, 43:1, and 78:1 that bracket the native TCGA-UT long tail. Using frozen Virchow2 patch features and WSI-bag features with the same 30-instance evidence budget as the companion benchmark, discrimination is evaluated for patch-feature and WSI-bag mitigation methods. Method rankings are preserved across imbalance severities, while balanced accuracy declines monotonically by three to four percentage points from $\lambda = 0.8$ to $\lambda = 1.3$.

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1 Introduction

The companion TCGA-UT imbalance benchmark compared class-imbalance mitigation methods in their native input regimes: labeled patch-feature classification and weakly supervised WSI-bag classification. It also showed that class frequency alone is an incomplete explanation for poor classwise performance. Some low-support cancer types remain separable under a frozen pathology foundation model, whereas other classes are difficult even when their support is not the smallest. This report turns that observation into an experimental design question: how should mitigation methods be interpreted when class frequency and inherent class difficulty can be varied deliberately?

TCGA-UT is a useful test bed for this question because it provides both slide labels and patch-level evidence derived from diagnostic whole-slide images in The Cancer Genome Atlas (Tomczak et al., 2015; Komura et al., 2022). The native dataset already has a long-tailed slide distribution over 32 cancer types. Native imbalance is realistic, but it entangles prevalence with many other factors: tumor morphology, tissue diversity, slide acquisition, and the representation geometry induced by the frozen encoder. A controlled construction complements the native benchmark by imposing known training-support profiles while holding validation and test data fixed.

The goal is not to introduce a new mitigation method. Instead, the report evaluates representative methods from the companion taxonomy under three constructed training distributions that vary only the imbalance severity while holding the class ranking fixed. Patch-level methods include cross-entropy, class weighting, balanced sampling, focal loss, and Scholz-style metric losses (Scholz et al., 2024). WSI-bag methods include class weighting, balanced sampling, focal loss, RankMix, SC-MIL, and MDE-MIL (Chen and Lu, 2023; Juyal et al., 2024; Ling et al., 2025). Patch and WSI-bag methods are reported separately because they consume different prediction units and answer regime-specific questions.

2 Dataset Construction

The construction starts from the same TCGA-UT case-disjoint split protocol as the companion benchmark. For each seed, the validation and test splits are kept fixed, and artificial imbalance is applied only to the training split. The split assignment is patient-disjoint: all slides and feature rows associated with one TCGA participant remain in exactly one of train, validation, or test. This isolates the effect of training support: model selection and held-out evaluation remain comparable across imbalance parameters and mitigation methods without patient leakage.

2.1 Full-Scale Training Pool

All 32 TCGA-UT cancer types are retained. Let $C = 32$ and let $\mathcal{D}_{\text{train}}$ denote the seed-specific training slide pool. For class c, let a_c be the number of available training slides before artificial resampling. The constructed training split must satisfy three constraints:

$$1 \le n_c \le a_c, \qquad \sum_{c=1}^{C} n_c = N, \qquad c \in \{1, \dots, C\}. \tag{1}$$

The first constraint keeps every class represented and prevents oversampling beyond available slides. The second fixes the training pool at N slides, shared across all imbalance parameters and seeds; the choice of N is discussed in Section 2.3. The third makes the benchmark a 32-class TCGA-UT study throughout.

After slide selection, the input evidence is capped in the same way as in the companion benchmark. Patch classifiers use up to 30 deterministic patches per slide. WSI-bag classifiers use up to 30 deterministic frozen feature instances per slide. In both regimes, Virchow2 features remain frozen (Vorontsov et al., 2024; Zimmermann et al., 2024). Features are read from the shared cls_patchmean Virchow2 store (concat(CLS, mean(patch tokens)), 2,560-dimensional vectors) used by the companion WSI-bag benchmark. The cap prevents slides with more extracted tumor patches from dominating the training objective while keeping the input budget aligned across regimes.

2.2 Power-Law Training Support

For an ordered class list $(c_1, \ldots, c_C)$, the intended share for rank i under imbalance parameter λ is

$$p_i(\lambda) = \frac{i^{-\lambda}}{\sum_{j=1}^{C} j^{-\lambda}}. \tag{2}$$

When $\lambda = 0$, the intended distribution is uniform over ranks. As λ increases, early-ranked classes receive more training support and late-ranked classes receive less. Raw targets $|\mathcal{D}_{\text{train}}|p_i(\lambda)$ are converted to integer counts by flooring and assigning residual slides by largest fractional remainder.

The target distribution is then projected onto the feasible set in Equation (1). Each class receives one guaranteed slide. Remaining slides are allocated according to Equation (2); classes whose requested support exceeds their available training support are capped, and the unused mass is redistributed over the uncapped classes using the same power-law ranks. This cap-and-redistribute procedure keeps the constructed split full-scale without dropping any cancer type.

2.3 Class Order and Training Pool

All constructed splits use the *native-prevalence order*: classes are ranked by their TCGA-UT training-split slide count, so that rank 1 is the most frequent cancer type and rank $C = 32$ is the least frequent. This ordering follows the realistic direction of the native TCGA-UT long tail.

Pool size. The choice of pool size N determines whether the cap-and-redistribute step in Section 2 is ever active. When $N = \sum_c a_c$ (the full available training pool), redistribution forces $n_c = a_c$ for every class regardless of λ, making the power-law parameter inert. To ensure that all three target λ values produce genuinely distinct distributions, N must be small enough that no class requires capping even at the steepest imbalance.

Define the maximum clean pool for a given λ and seed as

$$T^*(\lambda) = \min_c \frac{a_c}{p_c(\lambda)}. \tag{3}$$

For $N \le T^*(\lambda)$, the intended targets $N\,p_c(\lambda)$ are feasible for every class without capping. Setting $N = \lfloor \min_s T_s^*(1.3) \rfloor$, where the outer minimum is over the three seeds, guarantees that the steepest evaluated imbalance produces a capping-free power-law distribution on every seed. For the TCGA-UT training splits, this yields $N =$ 1,528 slides, approximately 25% of the full training pool.

Realized support profiles. The three λ values produce head-to-tail ratios of approximately 15:1 ($\lambda = 0.8$), 43:1 ($\lambda = 1.1$), and 78:1 ($\lambda = 1.3$), bracketing the native TCGA-UT ratio of approximately 28:1. The $\lambda = 0.8$ setting places the constructed distribution below native imbalance; $\lambda = 1.1$ is moderately more severe; and $\lambda = 1.3$ is the strongest regime evaluated. These three operating points verify whether mitigation methods provide useful protection as imbalance intensifies along a common class ranking.

3 Methods

The method taxonomy follows the companion benchmark, but this report uses it to study controlled training distributions rather than only the native TCGA-UT distribution. Cross-entropy is the no-mitigation reference. Data-level methods change minibatch exposure through balanced sampling or feature-space bag augmentation. Algorithm-level methods change the per-example loss through class weighting, focal loss, metric-derived objectives, or prototype-affinity objectives. WSI-specific methods change how slide bags are represented or mixed.

Regime	Family	Methods	Controlled question
Patch	Baseline and losses	CE, weighted CE, focal loss	Frequency weighting and hard-example emphasis.
Patch	Metric losses	CE + soft F1, CE + soft MCC	Macro-metric losses with balanced minibatch exposure.
Patch	Specialized	CFAL, divide-and-conquer	Prototype-affinity and expert fusion objectives.
WSI bag	Baseline and losses	Weighted CE, focal loss	Slide-level analogues of standard losses; weight_power=0 recovers plain MIL CE.
WSI bag	MIL-specific	Balanced sampling, RankMix, SC-MIL, MDE-MIL	Bag-level sampling, augmentation, contrastive structure, and expert training.

Table 1: Method groups evaluated under constructed TCGA-UT training distributions. Patch and WSI-bag methods are compared only within their own input regime.

Let y be the class label, $\mathbf{z} \in \mathbb{R}^C$ the logits, and p_y the softmax probability of the true class. Cross-entropy optimizes $-\log p_y$. Weighted cross-entropy multiplies this term by a normalized inverse-frequency weight. Focal loss uses

$$\mathcal{L}_{\text{focal}}(\mathbf{z}, y) = -(1 - p_y)^\gamma \log p_y, \tag{4}$$

so confidently classified examples contribute less to the gradient. The soft-F1 and soft-MCC variants add differentiable approximations of macro evaluation metrics to CE, following the imbalance-aware medical-image objectives of Scholz et al. (2024). RankMix, SC-MIL, and MDE-MIL are retained as WSI-bag methods because their mechanisms operate on bags rather than isolated patch embeddings (Chen and Lu, 2023; Juyal et al., 2024; Ling et al., 2025).

4 Evaluation Protocol

Each constructed split is indexed by seed and imbalance parameter $\lambda \in \{0.8, 1.1, 1.3\}$. For every method and constructed regime, hyperparameters are selected by validation macro F1 across three seeds; ties are broken by validation balanced accuracy. Selected configurations are evaluated on the fixed test split, and results are reported as mean $\pm$ standard deviation across seeds. Macro F1 is the primary discrimination metric because it weights cancer types equally. Balanced accuracy and accuracy are reported as complementary discrimination metrics.

The comparison is intentionally regime-specific. Patch-feature methods are compared only with other patch-feature methods, and WSI-bag methods are compared only with other WSI-bag methods. The two regimes share the case-disjoint splits, frozen Virchow2 representation family, and 30-instance evidence budget, but they differ in prediction unit and aggregation mechanism.

5 Results

The result analysis asks how methods respond as imbalance severity increases from $\lambda = 0.8$ to $\lambda = 1.3$ under the native-prevalence class order. Table 2 summarises the realized support profiles; Figure 1 shows their per-rank structure. Tables 3 and 4 compare mitigation methods within the patch-feature and WSI-bag regimes.

λ	Training slides	Support range	Head:tail ratio
0.8	1,528	18–268	≈15:1
1.1	1,528	10–425	≈43:1
1.3	1,528	7–543	≈78:1
native (full pool)	6,091	20–562	≈28:1

Table 2: Constructed training-support profiles under native-prevalence ordering. Support range and head:tail ratio are the minimum and maximum per-class slide counts averaged across seeds. The native full-pool row is shown for reference; it is not part of the constructed experiment.

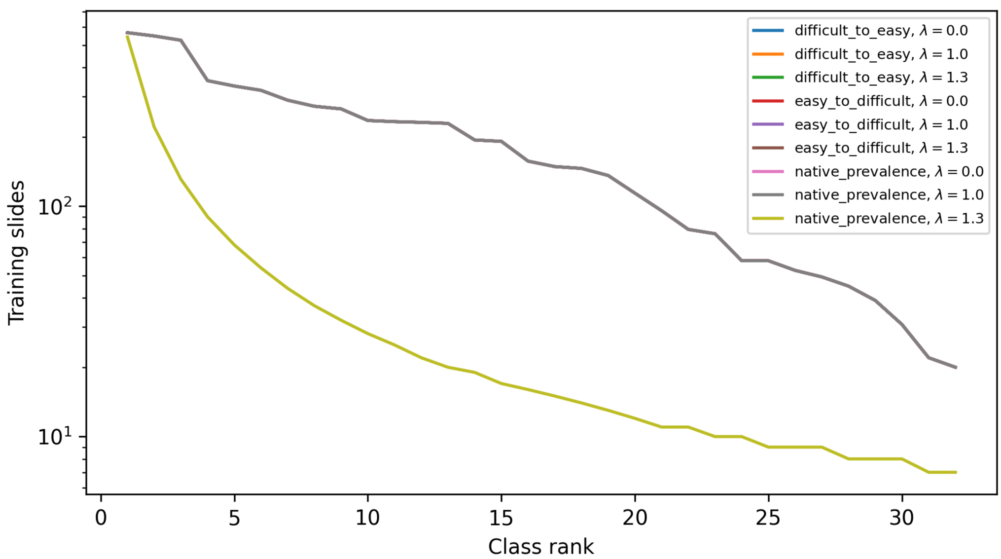

Figure 1: Constructed training support by class rank for the main imbalance settings.

Patch-feature results show a consistent, monotone decline as imbalance intensifies (Table 3). CFAL achieves the highest balanced accuracy at each severity: 0.721 at $\lambda = 0.8$, 0.701 at $\lambda = 1.1$, and 0.681 at $\lambda = 1.3$, a total decline of four percentage points. Focal loss and weighted CE are the next-strongest methods, each declining from approximately 0.718 to 0.672–0.679. Balanced sampling and the two metric-loss variants (CE+soft-F1 and CE+soft-MCC) follow closely, spanning 0.680–0.718 at $\lambda = 0.8$ and 0.670–0.680 at $\lambda = 1.3$. Divide-and-conquer is the weakest patch method and shows the steepest decline: from 0.676 at $\lambda = 0.8$ to 0.619 at $\lambda = 1.3$, a drop of nearly six percentage points. The method ranking is preserved at every severity; no method that is competitive at $\lambda = 0.8$ falls behind methods it outperformed there. The total variation within the patch method family at any fixed λ is approximately ten percentage points in balanced accuracy, a spread substantially larger than the three-to-four-point shift induced by moving from $\lambda = 0.8$ to $\lambda = 1.3$.

The WSI-bag regime shows a clearer separation between method families and a somewhat more complex imbalance response than the patch regime (Table 4). MDE-MIL achieves the

Method	Accuracy	Balanced accuracy	Macro F1
patch_feature_balanced_sampler_ce (native_prevalence, $\lambda = 0.8$)	0.765 ± 0.013	0.718 ± 0.008	0.696 ± 0.008
patch_feature_balanced_sampler_ce (native_prevalence, $\lambda = 1.1$)	0.761 ± 0.019	0.690 ± 0.024	0.684 ± 0.025
patch_feature_balanced_sampler_ce (native_prevalence, $\lambda = 1.3$)	0.750 ± 0.019	0.670 ± 0.009	0.672 ± 0.016
patch_feature_ce_soft_f1_balanced (native_prevalence, $\lambda = 0.8$)	0.759 ± 0.012	0.711 ± 0.006	0.682 ± 0.006
patch_feature_ce_soft_f1_balanced (native_prevalence, $\lambda = 1.1$)	0.754 ± 0.021	0.704 ± 0.013	0.680 ± 0.022
patch_feature_ce_soft_f1_balanced (native_prevalence, $\lambda = 1.3$)	0.744 ± 0.013	0.680 ± 0.004	0.668 ± 0.005
patch_feature_ce_soft_mcc_balanced (native_prevalence, $\lambda = 0.8$)	0.758 ± 0.011	0.707 ± 0.001	0.689 ± 0.005
patch_feature_ce_soft_mcc_balanced (native_prevalence, $\lambda = 1.1$)	0.753 ± 0.019	0.685 ± 0.019	0.681 ± 0.024
patch_feature_ce_soft_mcc_balanced (native_prevalence, $\lambda = 1.3$)	0.743 ± 0.015	0.670 ± 0.008	0.665 ± 0.008
patch_feature_cfal (native_prevalence, $\lambda = 0.8$)	0.775 ± 0.011	0.721 ± 0.009	0.703 ± 0.004
patch_feature_cfal (native_prevalence, $\lambda = 1.1$)	0.771 ± 0.016	0.701 ± 0.016	0.694 ± 0.018
patch_feature_cfal (native_prevalence, $\lambda = 1.3$)	0.761 ± 0.014	0.681 ± 0.009	0.679 ± 0.012
patch_feature_divide_conquer (native_prevalence, $\lambda = 0.8$)	0.741 ± 0.011	0.676 ± 0.012	0.672 ± 0.005
patch_feature_divide_conquer (native_prevalence, $\lambda = 1.1$)	0.741 ± 0.016	0.652 ± 0.020	0.661 ± 0.021
patch_feature_divide_conquer (native_prevalence, $\lambda = 1.3$)	0.724 ± 0.015	0.619 ± 0.007	0.633 ± 0.010
patch_feature_focal (native_prevalence, $\lambda = 0.8$)	0.769 ± 0.010	0.718 ± 0.004	0.698 ± 0.004
patch_feature_focal (native_prevalence, $\lambda = 1.1$)	0.768 ± 0.016	0.696 ± 0.015	0.690 ± 0.018
patch_feature_focal (native_prevalence, $\lambda = 1.3$)	0.754 ± 0.010	0.672 ± 0.006	0.670 ± 0.013
patch_feature_weighted_ce (native_prevalence, $\lambda = 0.8$)	0.771 ± 0.010	0.714 ± 0.004	0.698 ± 0.006
patch_feature_weighted_ce (native_prevalence, $\lambda = 1.1$)	0.768 ± 0.016	0.698 ± 0.014	0.690 ± 0.016
patch_feature_weighted_ce (native_prevalence, $\lambda = 1.3$)	0.759 ± 0.014	0.679 ± 0.007	0.677 ± 0.013

Table 3: Patch-feature test results for validation-selected configurations under constructed training distributions.

highest balanced accuracy at $\lambda = 0.8$ (0.830) and declines to 0.808 and 0.799 at $\lambda = 1.1$ and $\lambda = 1.3$. RankMix is the closest competitor and is notably stable: 0.830, 0.818, 0.820 across the three severities, with its performance at $\lambda = 1.3$ essentially matching $\lambda = 0.8$. SC-MIL follows at 0.822–0.807–0.787. Focal MIL declines more steeply: 0.816 at $\lambda = 0.8$ to 0.773 at $\lambda = 1.3$. Weighted MIL is the most stable loss-based method (0.799–0.791–0.798), with its $\lambda = 1.3$ result close to $\lambda = 0.8$. Balanced-sampling MIL occupies the lowest tier (0.793–0.795–0.769), with a more pronounced drop at the strongest imbalance. The ranking of method families—MDE-MIL and RankMix ahead of SC-MIL and focal MIL, ahead of loss-reweighting variants—is broadly maintained across all three severities, with only minor shuffling between focal MIL and MDE-MIL at $\lambda = 1.3$. The within-family advantage of bag-composition mechanisms over loss-based variants persists regardless of imbalance severity.

6 Discussion

The controlled construction reveals two complementary patterns. First, method rankings are preserved across imbalance severities: no method that is competitive at mild imbalance ($\lambda = 0.8$) falls behind methods it outperformed there as imbalance intensifies to $\lambda = 1.3$. Second, balanced accuracy declines monotonically with λ for almost all methods, with a total range of three to six percentage points from the mildest to the strongest constructed regime. These two patterns together suggest that the relative utility of mitigation methods is robust to the particular severity chosen within this range, but that absolute performance is sensitive to the support available for rare classes.

The monotone decline is more pronounced in the patch regime than the WSI-bag regime, and steeper for some methods than others. Divide-and-conquer falls six percentage points in balanced accuracy from $\lambda = 0.8$ to $\lambda = 1.3$, the largest decline of any patch method. CFAL and the loss-reweighting variants (weighted CE, focal loss) decline by approximately four percentage points. The metric-loss variants (CE+soft-F1, CE+soft-MCC) show a similar pattern but with somewhat larger seed variance at the strongest imbalance. The ten-point spread between the best (CFAL, $\approx$0.72) and worst (divide-and-conquer, $\approx$0.62) patch methods

Method	Accuracy	Balanced accuracy	Macro F1
mde_mil (native_prevalence, $\lambda = 0.8$)	0.854 ± 0.011	0.830 ± 0.023	0.811 ± 0.017
mde_mil (native_prevalence, $\lambda = 1.1$)	0.845 ± 0.017	0.808 ± 0.015	0.799 ± 0.017
mde_mil (native_prevalence, $\lambda = 1.3$)	0.844 ± 0.006	0.799 ± 0.004	0.794 ± 0.009
mil_balanced_sampler_ce (native_prevalence, $\lambda = 0.8$)	0.825 ± 0.018	0.793 ± 0.011	0.773 ± 0.012
mil_balanced_sampler_ce (native_prevalence, $\lambda = 1.1$)	0.823 ± 0.014	0.795 ± 0.014	0.782 ± 0.011
mil_balanced_sampler_ce (native_prevalence, $\lambda = 1.3$)	0.820 ± 0.004	0.769 ± 0.026	0.766 ± 0.014
mil_focal (native_prevalence, $\lambda = 0.8$)	0.845 ± 0.008	0.816 ± 0.002	0.796 ± 0.004
mil_focal (native_prevalence, $\lambda = 1.1$)	0.839 ± 0.011	0.800 ± 0.008	0.789 ± 0.012
mil_focal (native_prevalence, $\lambda = 1.3$)	0.830 ± 0.008	0.773 ± 0.011	0.778 ± 0.008
mil_weighted_ce (native_prevalence, $\lambda = 0.8$)	0.836 ± 0.009	0.799 ± 0.008	0.781 ± 0.009
mil_weighted_ce (native_prevalence, $\lambda = 1.1$)	0.835 ± 0.007	0.791 ± 0.007	0.780 ± 0.004
mil_weighted_ce (native_prevalence, $\lambda = 1.3$)	0.836 ± 0.010	0.798 ± 0.005	0.789 ± 0.010
rankmix_mil (native_prevalence, $\lambda = 0.8$)	0.848 ± 0.013	0.830 ± 0.020	0.797 ± 0.014
rankmix_mil (native_prevalence, $\lambda = 1.1$)	0.845 ± 0.016	0.818 ± 0.002	0.791 ± 0.005
rankmix_mil (native_prevalence, $\lambda = 1.3$)	0.849 ± 0.006	0.820 ± 0.012	0.800 ± 0.015
sc_mil (native_prevalence, $\lambda = 0.8$)	0.848 ± 0.006	0.822 ± 0.017	0.795 ± 0.016
sc_mil (native_prevalence, $\lambda = 1.1$)	0.840 ± 0.022	0.807 ± 0.012	0.793 ± 0.018
sc_mil (native_prevalence, $\lambda = 1.3$)	0.833 ± 0.009	0.787 ± 0.019	0.786 ± 0.012

Table 4: WSI-bag test results for validation-selected configurations under constructed training distributions.

at $\lambda = 1.3$ substantially exceeds the four-point shift induced by increasing λ, confirming that method choice matters more than exact severity in this range.

In the WSI-bag regime, RankMix and weighted MIL stand out for their stability across λ: both show less than two percentage points of decline from $\lambda = 0.8$ to $\lambda = 1.3$, suggesting that their mechanisms—bag-level mixing and inverse-frequency reweighting—respond differently to increasing class sparsity than aggregation-focused methods such as MDE-MIL and SC-MIL, which exhibit steeper declines. The ranking of bag-composition methods over loss-based variants is maintained throughout, indicating that the advantage of WSI-specific aggregation mechanisms is not contingent on the severity of training imbalance.

Taken together, these results extend the companion benchmark in a controlled direction. The companion demonstrated that class frequency alone does not predict classwise difficulty in the native TCGA-UT distribution. The present experiment shows that when frequency is imposed directly via a clean power-law construction, aggregate discrimination declines predictably but modestly with imbalance severity, and the method ranking remains informative across the range tested. The prediction unit—patch versus WSI-bag—continues to be the dominant determinant of absolute performance: WSI-bag methods achieve balanced accuracy twelve to fifteen percentage points higher than their patch-feature counterparts at equivalent imbalance.

7 Conclusion

This report defines a full-scale controlled TCGA-UT imbalance benchmark that keeps all 32 cancer types, preserves patient-disjoint validation and test splits, and varies the training support profile under native-prevalence class ordering at imbalance parameters $\lambda \in \{0.8, 1.1, 1.3\}$. A fixed pool of 1,528 slides ensures that the power-law construction remains capping-free at all three severities, yielding realized head-to-tail ratios of approximately 15:1, 43:1, and 78:1 that bracket the native TCGA-UT long tail. The primary finding across all three constructed regimes is that method rankings are preserved while balanced accuracy declines monotonically by three to six percentage points from $\lambda = 0.8$ to $\lambda = 1.3$. At the patch level, CFAL achieves the highest balanced accuracy and shows the most controlled decline; divide-and-conquer is the weakest patch method and suffers the steepest loss. At the WSI-bag level, MDE-MIL and

RankMix lead across all severities; RankMix and weighted MIL are the most stable methods under increasing imbalance. The preserved ranking across severities indicates that method selection can be informed by experiments at mild imbalance, while the monotone decline confirms that stronger imbalance meaningfully degrades absolute performance. The prediction unit remains the dominant determinant: WSI-bag methods achieve balanced accuracy roughly fifteen percentage points above patch-feature methods at equivalent imbalance severity.

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